CarrySelect-4: A Registered 4-bit Carry-Select Adder from RTL to GDSII in SKY130 with a Logical-Effort Study of a Full-Custom Mirror Full Adder

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***Abstract*—We present CarrySelect-4, a 4-bit carry-select adder (CSLA) wrapped in a valid/ready interface and taken through a complete open-source flow on the SkyWater SKY130 process. The datapath splits the adder into a 2-bit ripple block and two speculative 2-bit blocks whose results are selected by the low carry; a three-state Moore controller registers inputs and outputs, giving a latency of 2 cycles and one result every 2 cycles. Exhaustive self-checking simulation passes 512 of 512 input vectors. Synthesis with Yosys uses 218 NAND2 equivalents, and LibreLane places and routes the design at 100 MHz with positive setup and hold slack in all nine signoff corners and zero DRC and LVS errors. In a second branch we size, lay out and extract a 28-transistor mirror full adder in Magic and compare a Logical Effort estimate of the ripple-carry path with pre- and post-layout SPICE: the estimate of 312 ps lies between the schematic result (264 ps) and the extracted result (420 ps).**

***Index Terms*—carry-select adder, SKY130, OpenLane, LibreLane, logical effort, full-custom layout, mirror adder**

I. Introduction

The carry chain sets the delay of an n-bit ripple-carry adder (RCA), which grows as $O(n)$. A carry-select adder [1] computes the upper bits twice, once for each value of the incoming carry, and picks the right result with a multiplexer, trading area for delay [2]. This work implements a small but complete instance of that idea and follows it through every step of a modern open-source flow: RTL and verification, synthesis and static timing with Yosys [3] and OpenSTA [4], and physical design with OpenLane 2 / LibreLane [5], [6] on SKY130 [7]. We then study the adder's leaf cell, a mirror full adder, at transistor level with Logical Effort [8] and Magic [9], and check how well the hand estimate predicts SPICE before and after layout.

II. Architecture

Fig. 1 shows the top level. The 9-bit input word $\{c_{in}, b, a\}$ is captured in an input bank (9 FF) when `ld_en = valid_in & ready_out`. The CSLA uses six full adders in three 2-bit RCAs: *low* adds $a[1{:}0] + b[1{:}0] + c_{in}$, while *hi0* and *hi1* add $a[3{:}2] + b[3{:}2]$ with a carry-in of 0 and 1. The low carry c_2 drives the select inputs of three 2:1 multiplexers for `sum[3:2]` and `cout`. Full adder and multiplexer modules

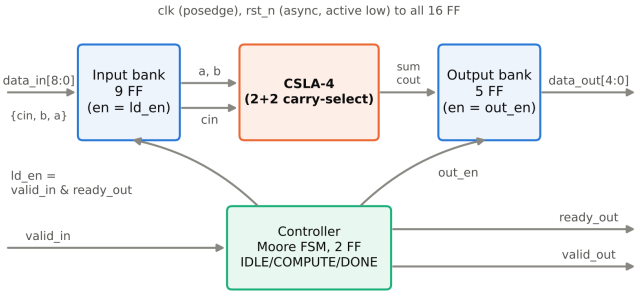


Fig. 1. Registered CarrySelect-4 datapath and controller.

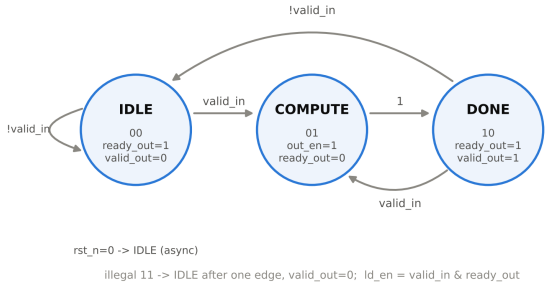


Fig. 2. Moore controller: state codes, outputs and transitions.

keep their hierarchy during synthesis so that the carry-select structure survives technology mapping. The 5-bit result is stored in an output bank (5 FF). A Moore controller with two state bits (Fig. 2) sequences IDLE, COMPUTE and DONE; the unused code returns to IDLE. All 16 flip-flops use an asynchronous active-low reset.

III. Verification

A self-checking SystemVerilog testbench runs on Verilator and compares every output against vectors from an independent Python golden model. Table I lists the cases. Fig. 3 shows back-to-back transfers: an input accepted at clock edge

TABLE I
VERIFICATION CASES (VERILATOR, SELF-CHECKING)

Case	Scenario	Result
1	directed vectors after reset	PASS
2	back-to-back, latency and throughput	PASS
3	random `valid_in` bubbles (30%)	PASS
4	async reset in COMPUTE and in DONE	PASS
5	exhaustive, 512 vectors, 0 errors	PASS
–	illegal state `2'b11`	PASS

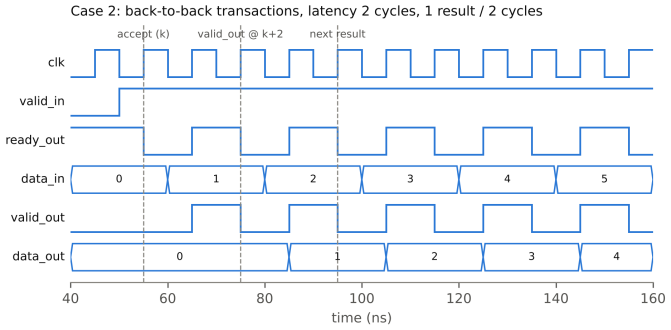


Fig. 3. Case 2 waveform: latency of two cycles, one result every two cycles.

k produces `valid_out` at edge $k+2$, and the next result follows two cycles later.

IV. Synthesis and Physical Design

Yosys maps the design to `sky130_fd_sc_hd` cells: 49 cells, 817 μm^2, i.e. 218 NAND2 equivalents against a budget of 500. With a 10 ns clock and 2 ns input/output delays, OpenSTA reports a worst setup slack of 6.78 ns at the typical corner; the register-to-register critical path `a_q[0] -> cout_q` arrives after 1.78 ns.

Fig. 4 compares the CSLA with a 4-bit RCA. With hierarchy kept, the CSLA is faster (1.24 vs. 1.71 ns) but larger (236 vs. 135 μm^2), so its area–delay product is worse. When ABC may flatten the logic, both shrink and slow down; the flat CSLA keeps the lower delay (1.59 vs. 2 ns) and gives the best area–delay product. At four bits the duplicated upper block costs nearly as much as it saves.

LibreLane places and routes the design with a 55% core-utilization target and aspect ratio 1 (Fig. 5). Table II summarizes signoff. Setup and hold slack are positive in all nine corners (min/nom/max extraction $\times$ ss/tt/ff), including the recovery/removal checks of `rst_n`. The estimated maximum frequency at the typical corner is 277 MHz. A second run at a 5 ns clock closes at the typical and fast corners but misses setup at the slow corner by -0.045 ns.

V. Full-Custom Full Adder and Logical Effort

The leaf cell is a 28-transistor mirror adder: a majority (carry) gate produces $\overline{C_o}$, a sum gate reuses it to produce $\overline{S}$, and two inverters restore polarity. An FO4 inverter simulation in ngspice (typical corner) gives $\tau = 12.9$ ps (FO4 $= 64.5$ ps) and $p_{inv} = 1.13$ for W_p/W_n = 0.84/0.42.

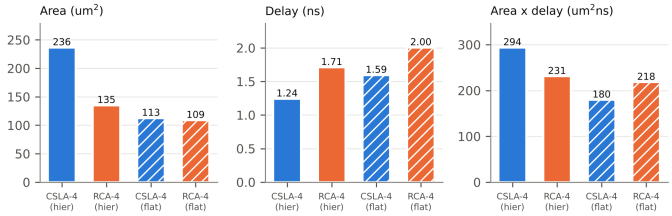


Fig. 4. CSLA-4 vs. RCA-4 after synthesis (typical corner); hatched bars are flattened netlists.

TABLE II
POST-ROUTE SIGNOFF AT 100 MHZ

Metric	Value
Die area (µm^2)	3420
Std cells / final utilization (%)	137 / 81
Setup slack, tt / worst corner (ns)	6.39 / 4.8
Hold slack, tt / worst corner (ns)	0.327 / 0.111
f_{max} estimate, tt (MHz)	277
DRC Magic / KLayout, LVS errors	0 / 0 / 0
Antenna violations	0
Total power (mW), wirelength (µm)	0.342, 2200

We analyse the 2-bit ripple path inside each RCA block, from a_0 through two carry gates and two inverters to c_2, which drives three multiplexer select pins ($C_{load} = 3\,C_{S,\mathrm{mux2}}$). With $g_{carry} = 2$, $g_{inv} = 1$ and branching taken from the transistor widths,

$$F = GBH, \qquad \hat{f} = F^{1/N}, \qquad D = N\hat{f} + P, \tag{1}$$

where $G = 4$, $B = 31.5$, $H = 1.45$ and $N = 4$, so $F = 183$ and $\hat{f} = 3.68$. With $P = 2p_{carry} + 2p_{inv}$ and a simulated $p_{carry} = 3.6$, $D = 24.2\,\tau = 312$ ps. Sizing S1 scales each transistor by its stack depth; sizing S2 also gives the carry-out inverter the size that Logical Effort assigns to the last stage. S2 is faster in SPICE (264 vs. 306 ps) and was laid out.

The layout (Fig. 6) orders the shared polysilicon columns along Euler paths, so the pMOS and nMOS rows need a single diffusion break. Full-width n- and p-tap rows under the supply rails keep every device within a few micrometres of a tap, far inside the 15 μm latch-up limit. The cell measures 16.2 $\times$ 12.3 μm (200 μm^2), is DRC-clean with the full SKY130 deck, and matches the schematic in Netgen LVS. Table III and Fig. 7 compare the estimate with SPICE. Relative to the schematic simulation the estimate is off by 18.2% (too slow); relative to the extracted netlist it is off by -25.7% (too fast). The schematic omits source/drain junction capacitance, and extraction adds it together with 80 wiring capacitors; Logical Effort folds all of this into a single parasitic term per stage.

VI. Conclusion

CarrySelect-4 meets its functional and timing targets: exhaustive simulation passes, the routed design closes at 100 MHz in every corner with clean DRC, LVS and antenna checks, and it uses less than half of its area budget. At four bits the carry-select structure is faster than ripple carry but

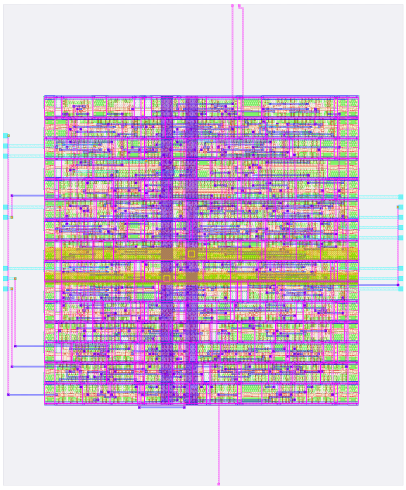


Fig. 5. Routed layout of `carryselect4_top` (LibreLane, SKY130).

TABLE III
RIPPLE-CARRY PATH DELAY $a_0 \rightarrow c_2$

Method	Delay (ps)	LE error
Logical Effort	312	–
SPICE, schematic	264	18.2%
SPICE, extracted	420	-25.7%

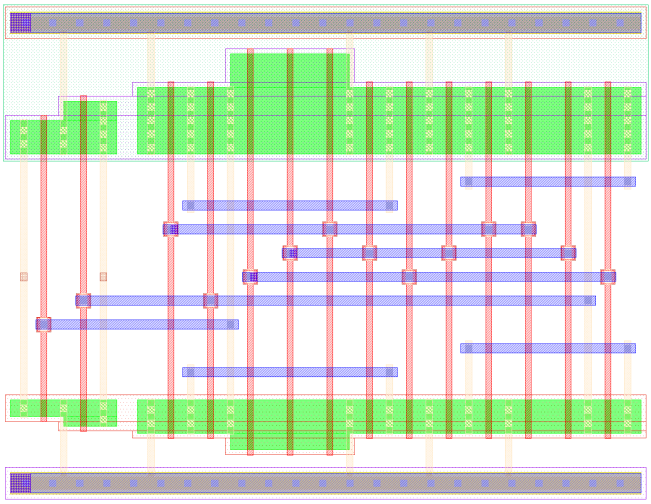


Fig. 6. Full-custom 28T mirror full adder in Magic (DRC and LVS clean).

does not win on area–delay product unless synthesis is allowed to flatten it. The full-custom study shows that Logical Effort picks the faster sizing and lands within the pre/post-layout band, but a layout-aware parasitic term is needed for accurate absolute delay. The custom cell is a study cell and is not used in the standard-cell flow.

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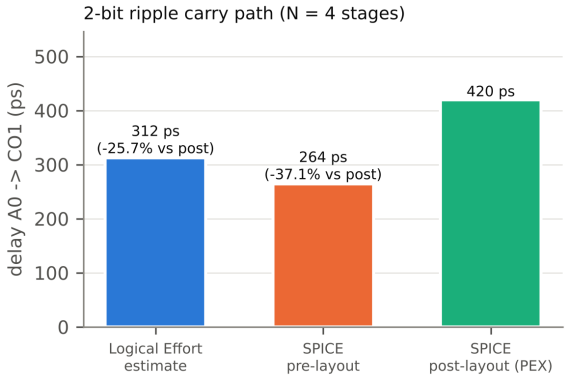


Fig. 7. Logical Effort estimate vs. SPICE before and after extraction.